

Distribution and upper bound of mimic numbers

OPG-36961	L1 - Tractable	Partiellement résolu	Number Theory
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Problem

Let the notation $a|b$ denote " a divides b ". The mimic function in number theory is defined as follows [1].

Definition For any positive integer $N = \sum_{i=0}^n \mathcal{X}_i \mathcal{M}^i$ divisible by $\mathcal{D}$, the mimic function, $f(\mathcal{D} | N)$, is given by,

$$f(\mathcal{D} | N) = \sum_{i=0}^n \mathcal{X}_i (\mathcal{M} - \mathcal{D})^i$$

By using this definition of mimic function, the mimic number of any non-prime integer is defined as follows [1].

Definition The number m is defined to be the mimic number of any positive integer $N = \sum_{i=0}^n \mathcal{X}_i \mathcal{M}^i$, with respect to $\mathcal{D}$, for the minimum value of which $f^m(\mathcal{D} | N) = \mathcal{D}$.

Given these two definitions and a positive integer $\mathcal{D}$, find the distribution of mimic numbers of those numbers divisible by $\mathcal{D}$.

Again, find whether there is an upper bound of mimic numbers for a set of numbers divisible by any fixed positive integer $\mathcal{D}$.

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